# MealSenseAI Project Proposal

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A photo-based meal logger that recognises foods, estimates nutrition and explains every result.

## Problem

Logging meals by hand is tedious, and people managing diabetes or their weight need numbers they can trust and understand. This proposal was submitted on 20 June 2026.

Generic food apps rarely recognise local dishes and rarely say how sure they are.

## Proposed solution

MealSenseAI takes a photo, performs Food Recognition on each item on the plate, and turns portions into Nutrition Estimation: calories, carbohydrate, protein, fat, fibre and sodium. Every estimate is shown with its confidence and the alternatives the model considered, which is the Explainable AI requirement.

For users who manage their blood sugar, the app adds an educational view of Glycaemic Response: how a meal's carbohydrate and glycaemic load might affect glucose after eating.

## Safety

The app never gives medication or insulin advice, labels every estimate as an estimate, and uses health information only with explicit consent. Dietary guidance follows the Health Promotion Board's My Healthy Plate.

Illustrative sample document for the CEREBRO demo library.